

3(a)	<i>three from:</i> 1 <i>idea of</i> halving the number of chromosomes ; 2 (to make) haploid / n, gametes ; 3 (so) on fertilisation, (restores) diploid number / 2n ; 4 maintains constant chromosome number / prevent chromosome number doubling ;	3
3(b)(i)	<i>two from:</i> 1 ref. to light intensity and change in size of, plant / part of plant ; 2 wind speed and size of plant ; 3 waterlogged soil and leaves / flowers, above water ; 4 pH and flower colour ; 5 AVP ; environmental condition and effect 6 AVP ; environmental condition and effect	2
3(b)(ii)	<i>two from:</i> 1 genetic variation / new alleles ; 2 (therefore) new phenotypes ; 3 (new phenotype) has a selective advantage / is selected for ; 4 more likely to survive (selection pressure) ;	2

Question	Answer	Marks
3(c)	<i>two from:</i> 1 in full nutrient there is little difference (between two populations) ; 2 in low phosphate non-functional X root growth is, longer / ora ; 3 in low phosphate both populations grow less well than in full nutrient / ora ; 4 comparative data quote to support mp1 / mp2 / mp3 ; A processed data	2
3(d)(i)	69.67 to 70.6 ; ; <i>max 2 d.p</i> allow 1 working mark for $21 - 33 = (-)12$ (for calculating difference between means) $\sqrt{0.5^2 / 30 + 0.8^2 / 30}$ (for calculating the standard error in the difference)	2
3(d)(ii)	<i>reject null hypothesis</i> $t / 69.67$, is higher than, 2.00 / the critical value (at $p = 0.05$) ; there is a significant difference between, the means of the functional X and non-functional X populations ; allow ecf from calculated value of t	2